Software Engineering

Spring 2026

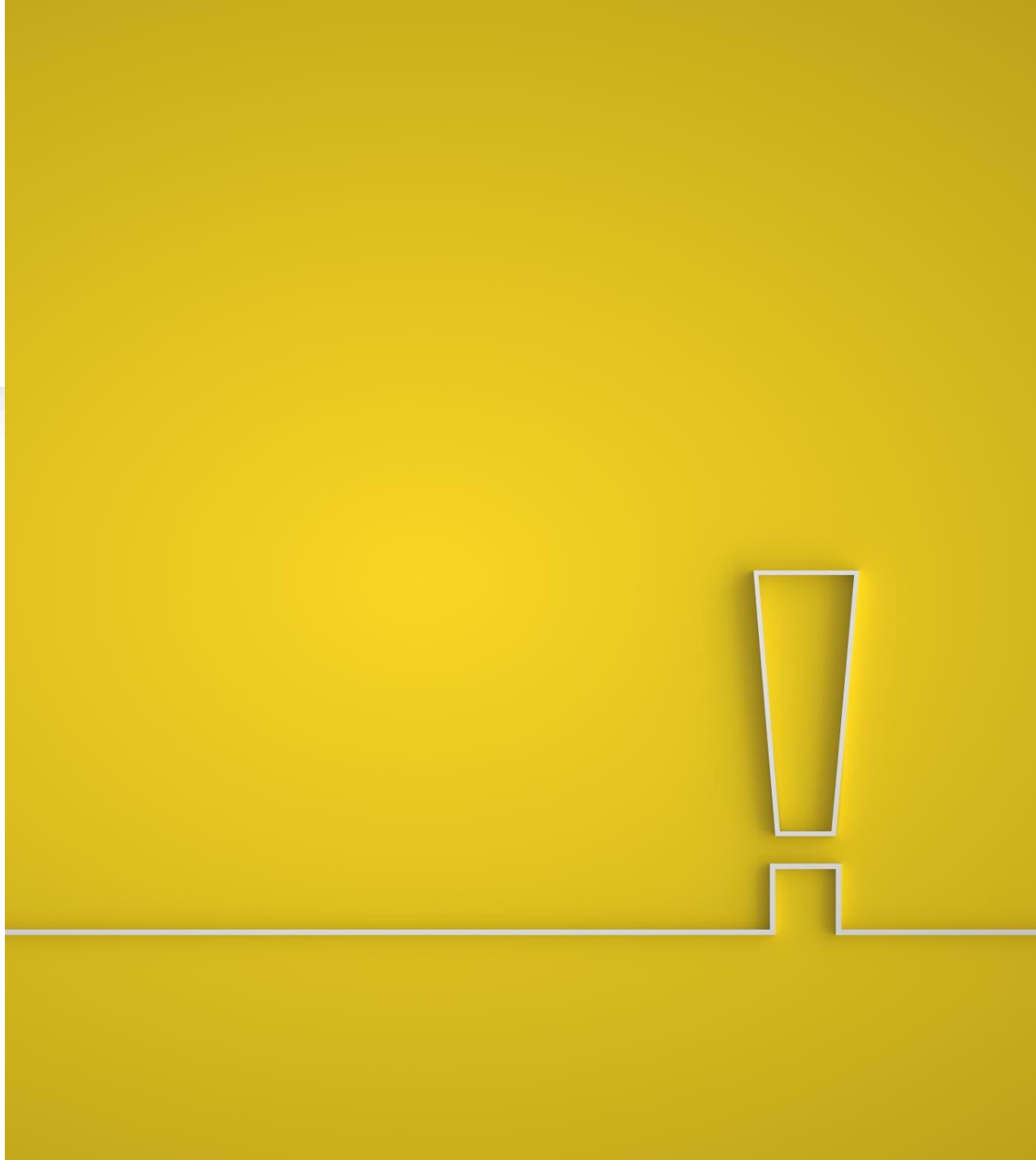
Henry
VILLAR

INTRODUCTION & SE MOTIVATION

SE Syllabus, Topics, Policies, Rules, Organization and Grading + Team Project and M0

Acknowledge

- This is shared content that the instructors of this course from the different sessions are using. Slides are based on course materials created by Dr Dragutin Petkovic and updated by Henry Villar
- There are 4 Professors teaching SE:
  - [01] Isabel Hyo Jung Song
  - [02] Henry Villar (Online Synchronous)
    - Our Grader [02]: Jaideep Reddy Gade.
  - [03] Dragutin Petkovic
  - [04] Isabel Hyo Jung Song
  - [05] Jose Ortiz Costa
  - Anthony John Souza [TA – General CSC 648-848]



Agenda

- **Introductions and SE Motivation** – Why is this class important?
- **About the class:** what will be covered; final class project
  - **Student responsibilities**, attendance, participation, team collaboration
  - **Canvas** site overview – will contain class material
  - **Policies & Grading**
  - Class **team assignments**
- **Team Project**
- **Milestone 0**
- **Roll Call** / Enrollment and pre-requisites review.
- Office hours (Tuesdays After class 8:45 PM – 9:15 PM – Please send an email)

Spring 2026 Calendar* – CSC 648-848 [02]

*Subject to change

Week	Date	Topic	Team Project	Points
W1	January 27	Introduction and Motivation for Software Engineering	Team Assignments	
W2	February 3	Requirements and Specifications: Personas & Use Cases	*Configuration Chosen (IT Infra & SW Stack)*	
W3	February 10	Software Engineering Teamwork & Arch Intro	**Milestone 0:** IT Infra & SW Stack + Individual video	10
W4	February 17	Software Architectures and Design		
W5	February 24	Design: Usability and GUI Principles, Design, Development and Evaluation	**Milestone 1:** High Level Requirements & Specifications, Use Cases and Architecture	*(10)*
W6	March 3	Software Project Management, SE and AI Ethics, IP Issues		
W7	March 10	Implementation: Coding +Style and Coding Practices	**Milestone 2:** More Detailed Requirements, Specs, Architecture, UI mock-ups and Vertical SW prototype	*(5)*
W8	March 17	Security Engineering	**Vertical Prototype**	*(5)*
	March 24 |March 31	Spring Recess – March 23-27 | March 31st Cesar Chavez Day		
W9	April 7	Validation: SW Testing & QA	**Milestone 3:** Review of functionality, UI, SW and planning for final product delivery	*(10)*
W10	April 14	Cont. Validation: Usability Testing and Focus Groups		
W11	April 21	Software Engineering: Methodology & Process	**Milestone 4:** Beta Launch, QA and Usability Testing and Final Commitment for Product Features (P1 list).	*(10)*
W12	April 28	Software Engineering: Preparing for the Future. Best Practices and What makes a great Software Engineer?		
W13	May 5th	Software Engineering Review. Prep for M5 Demo & Final Exam	Talk to Team leads	
W14	May 12	Final Project Demo – SW Release	**Milestone 5:** Final Demo and Documents for Grading (10)	50
W15	May 19	FINAL EXAM		40

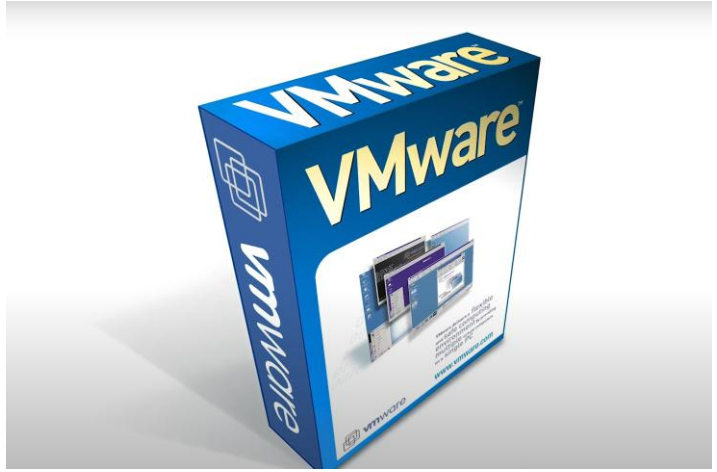
Team Assignments

Last Name	First Name	Email	Team
Kumar	Mohit	mkumar3@sfsu.edu	1
Naranja	John Carlo	jnaranja@sfsu.edu	1
Sengchareune	Khaterina	ksengchareune@sfsu.edu	1
Hernandez	Alex	ahernandez75@sfsu.edu	1
Brust	Will	wbrust@sfsu.edu	1
Weerasuriya	Lasiru	lweerasuriya@sfsu.edu	1
Harish	Shreya	sharish@sfsu.edu	1
Sanchez	Brandon	bsanchez15@sfsu.edu	2
Wong	Laura	lwong25@sfsu.edu	2
Berhane	Mikias	mberhane1@sfsu.edu	2
Lubrino	Cielina Maree	clubrino@sfsu.edu	2
Le	Jason	jle34@sfsu.edu	2
Nadery	Omeid	onadery@sfsu.edu	2
Perez	Damian	dperez52@sfsu.edu	2
Romero	Martin	mromero13@mail.sfsu.edu	3
Velazquez Aguilera	Yanitsy	yvelazquezaguilera@sfsu.edu	3
Chen	Richie	lchen22@sfsu.edu	3
Teves	Kaylen	kteves@sfsu.edu	3
Rodriguez-Malak	Carlos	crodriguezmalak@sfsu.edu	3
Barra Avila	Franco	fbarraavila@sfsu.edu	3
Cancino	Cj	ccancino@sfsu.edu	3
Bhatta Lama	Sargam	sbhattalama@sfsu.edu	4
Ashby	Kaitlyn	kashby@sfsu.edu	4
Jezewski	Houdini	hjezewski@sfsu.edu	4
Torres	John	jtorres41@sfsu.edu	4
Wright	Emma	ewright12@sfsu.edu	4
Robles	Jacob	jrobles4@sfsu.edu	4
Qazi	Saeed	sqazi@sfsu.edu	4
Li	Mark	cli36@sfsu.edu	5
Jiang	Yuejun	yjiang13@sfsu.edu	5
Yu	Kerry	kyu19@sfsu.edu	5
Wahid	Mahvesh	mwahid@sfsu.edu	5
Kligman	Ian	ikligman@sfsu.edu	5
Kooner	Reman	rkooner@sfsu.edu	5
Caya	John Peter	jcaya@sfsu.edu	6
Rubio	Christopher	crubio4@sfsu.edu	6
Pasami	Gabriella	gpasami@sfsu.edu	6
Houlston	Foster	fhoulston@sfsu.edu	6
Young	Will	wyoung@sfsu.edu	6
Pokharel	Poojan	ppokharel1@sfsu.edu	6

SAN FRANCISCO STATE UNIVERSITY

Today's Outcomes

1. Understand exactly the expectations of the **Software Engineering** class: Project-based.
2. Team familiarity – Assignments
    1. Roles & Responsibilities
3. Talk about the Infrastructure and Software Stack
4. Final Team Project
5. Milestone 0



Final Objective, the Pragmatic One

About Me: JEVP, Henry Villar @Splunk/Cisco Previously: VMware, NetApp & Intel

- EDUCATION

- *Liberty University*
- ***Ph.D.*** *Developmental Psychology. Student (2022-Present)*

- *San Francisco State University*
- ***Masters*** *of Science in Computer Science. Computing and Business (2011)*
- ***Bachelor*** *of Science in Computer Science, and Minor in Mathematics (2006)*

- *Naval Academy of Mexico*
- *Bachelor of Science in Hydrographic* ***Engineering****. Valedictorian (1993)*

Students Introductions

YOUR NAME, LOCATION & PREFERRED PROGRAMING LANGUAGE

ARE YOU ALREADY WORKING? TECH OR NOT TECH?

SOMETHING ABOUT YOU THAT WE CAN'T TELL JUST BY LOOKING AT YOU ;^)

Motivation

Software Engineering (SE) is “The **application** of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software”.

"Software Engineering 2004": **Curriculum** Guidelines for Undergraduate Degree Programs in Software Engineering, The joint Task Force on Computing Curricula, IEEE Computer Society, Association for Computing Machinery, August 2004

Motivation

- Most of the **failures** of delivering SW to specs, on time and budget, and to a user "delight" were rooted in a *misunderstanding user needs, and/or poor design, planning and organization.*

Charette, R. N., "Why Software Fails", IEEE Spectrum, September 2005, pp. 42

Motivation

- ACM study on SW outsourcing and globalization specifically mentions the need to train SE students in so called "**soft skills**", *teamwork, communication, SE process organization and issues related to globalization*

Aspray W., Mayadas F., Vardi M.Y., Editors: "Globalization and Offshoring of Software, A Report of the ACM Job Migration Task Force", ACM 2006, http://www.acm.org/globalizationreport/

Motivation

- Check job postings for SW engineers – most require (besides specific technical skills):
  - Ability to work in a team
  - Oral and written communication
  - Knowledge of SE tools
  - Familiarity with modern SE processes and lifecycle (NEW emphasis on continuous and collaborative SE)
  - Awareness about user experience

- Numerous student testimonials how the class helped them find the job or in current job

Some things never change…

SW Technology (tools, frameworks, languages, components, protocols) do change very often (yearly…)

BUT

Principles and best practices of SE management, organization and people stuff change little with time

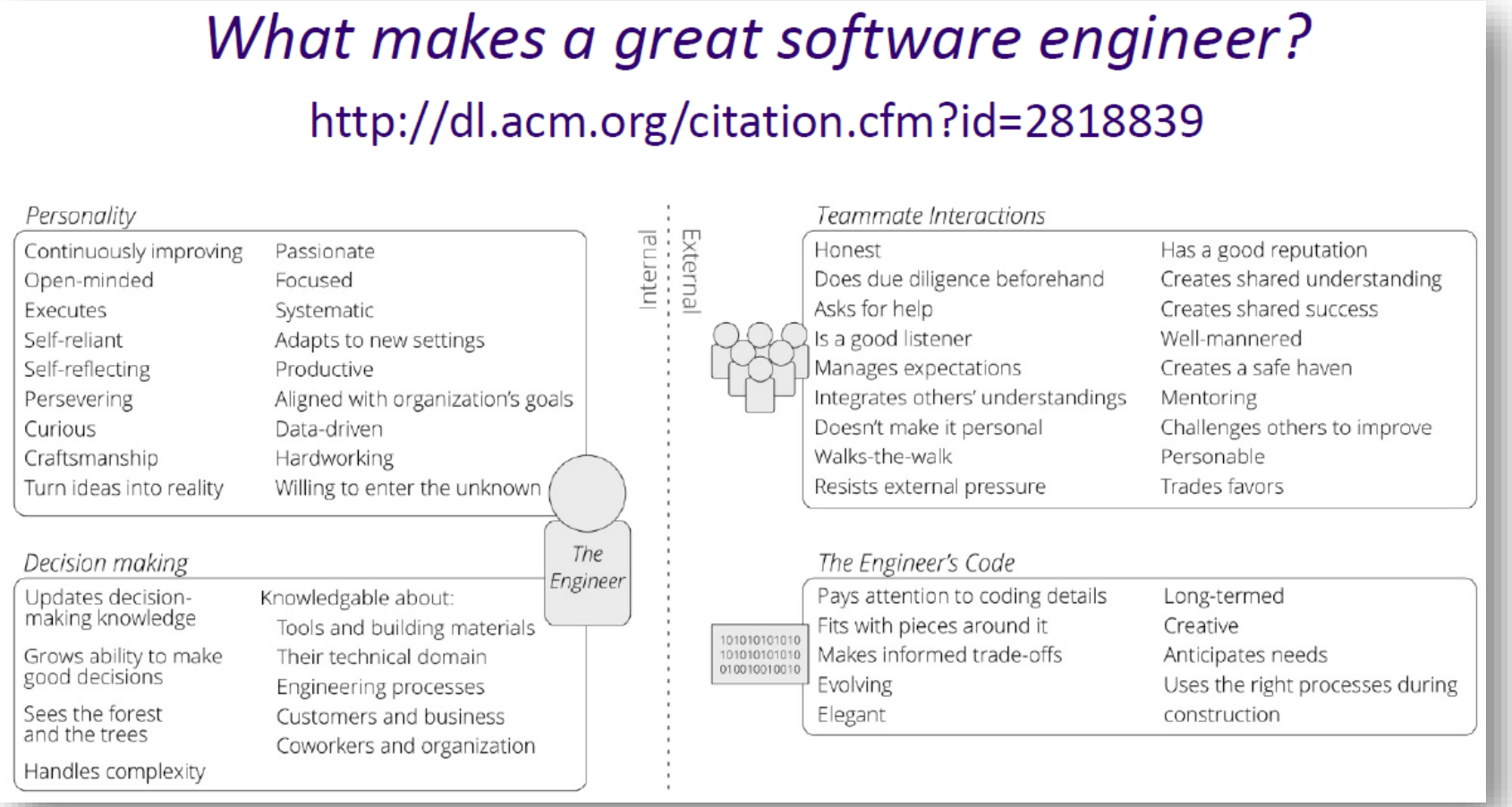
Hence, learning SE principles (from this class) offers you long term skills which grow in importance as you get more experienced

Hard (technical) and soft (interpersonal/professional) skills for SE

- Hard skills – traditional CS curriculum
  - Data structures, Math
  - Programming
  - Computer architectures
  - Database/networking
  - ….
- BUT ALSO: Soft skills
  - Teamwork and communications
  - Ability to follow SE best practices and processes
  - Dealing with customers, stakeholders, management
  - SW development using team tools

What makes great SW Engineer and how to teach SW Engineering

Check ACM slides posted on Canvas



Science vs. Engineering

- Grow and expand knowledge
- "Build in order to learn"
- "Vertical" education – go into depth and details

- Emphasis on individual competency

- Apply knowledge to build useful products
- "Learn in order to build"
- "Horizontal" education – cover all aspects to some degree – via a project from inception to delivery
- Emphasis on teamwork

What makes a great SW engineer -0 check ACM slides (posted)

About CSC 648/848

Class goals: make you a better SW Engineer and give you skills to be successful in the marketplace

This class is NOT about coding (but is about programming practices, application and style)

This class is going to help you learn how to work in a team to build SW on time, budget and per specs which delights the customers

You will learn:

- Valuable best practices and methods for successful SE (continuous collaborative development, teamwork) and get experience close to real working environment – great for those looking or work but never worked before
- Modern SW development tools

You should update your Resume/CV after class – instructions will be posted

Class experience and final project portfolio are great for job search

Class teaching philosophy

Teach *SE best practices, concepts and terms* – focus on practical SE

A lot of class discussion and real-life examples

In parallel let student teams implement above best practices around team project

Teach what they need when they need it

Immerse, supervise and coach student teams (6-7 students) in as realistic setting as possible (model industry startup setting)

Above methods:

Ensures real learning "by doing it"

More fun, less boring

Better learning

Teaching of SE "theory" and best practices is synchronized and driven by milestones

Topics

4) GUI, usability

7) Des. Patterns, Coding style, Src. Cntrl,

3) Req&Specs

6) PM, teamwork

9) Guest speakers

11) SW quality

5) UML, SW processes

2) Final project

8) QA usability

10) ethics, legalese

Time

M0

M1
high
level specs

M2
detailed
specs

M3
prototype
and arch.
Final proj.
specs

M4
beta
and usability

M5
final
delivery
and demo

Milestones

Spring 2025 Calendar* – CSC 648-848 [02]

*Subject to change

Week	Date	Topic	Team Project	Points
W1	January 27	Introduction and Motivation for Software Engineering	Team Assignments	
W2	February 3	Requirements and Specifications: Personas & Use Cases	*Configuration Chosen (IT Infra & SW Stack)*	
W3	February 10	Software Engineering Teamwork & Arch Intro	**Milestone 0:** IT Infra & SW Stack + Individual video	10
W4	February 17	Software Architectures and Design		
W5	February 24	Design: Usability and GUI Principles, Design, Development and Evaluation	**Milestone 1:** High Level Requirements & Specifications, Use Cases and Architecture	*(10)*
W6	March 3	Software Project Management, SE and AI Ethics, IP Issues		
W7	March 10	Implementation: Coding +Style and Coding Practices	**Milestone 2:** More Detailed Requirements, Specs, Architecture, UI mock-ups and Vertical SW prototype	*(5)*
W8	March 17	Security Engineering	**Vertical Prototype**	*(5)*
	March 24 |March 31	Spring Recess – March 23-27 | March 31st Cesar Chavez Day		
W9	April 7	Validation: SW Testing & QA	**Milestone 3:** Review of functionality, UI, SW and planning for final product delivery	*(10)*
W10	April 14	Cont. Validation: Usability Testing and Focus Groups		
W11	April 21	Software Engineering: Methodology & Process	**Milestone 4:** Beta Launch, QA and Usability Testing and Final Commitment for Product Features (P1 list).	*(10)*
W12	April 28	Software Engineering: Preparing for the Future. Best Practices and What makes a great Software Engineer?		
W13	May 5th	Software Engineering Review. Prep for M5 Demo & Final Exam	Talk to Team leads	
W14	May 12	Final Project Demo – SW Release	**Milestone 5:** Final Demo and Documents for Grading (10)	50
W15	May 19	FINAL EXAM		40

About the class organization

- Class structure (unless otherwise):
  - First half of the class – teaching and discussion (Lecture)
  - Second half of the class – supervised teamwork ***MANDATORY attendance ***
- Milestone 0 — learn SE tools and each team organise their IT (with our supervision)
- Final Team Project: simulating small SW company: Students will work in teams of 5-7 to produce a SW product (same for all teams). Each student team will deliver 5 group Milestones and demo their project at the end.
  - High level final team project description to be posted on Canvas – same for all teams
- Final individual exam – tests basic knowledge
- No formal homework, no midterm BUT a lot of teamwork and 6 formal milestones (M0, M1-M5) an final individual exam

- Teamwork participation is the key and mandatory!!!!

Mandatory class policies

- Attendance first 4 weeks full class is mandatory, so students can start strong working with their teams and establish teamwork – if not assigned to the team student will be dropped.
- Fifth week and later:
  - Must attend teamwork session (last 1+ hours of the class)
- All Teams will present TAKEAWAYS from previous class
- Must participate in team activities and follow good team ethics and behavior.
- Must read and respond to team and instructor e-mails within 24 hours
- Must follow the requested deadlines
- Must use class mandated SE and communication tools for final project development
- Must report absence to team and instructors ahead of time to instructor and team lead

Failure to abide by class policies will result in reduced grade points

Communication Policies: instructors - students

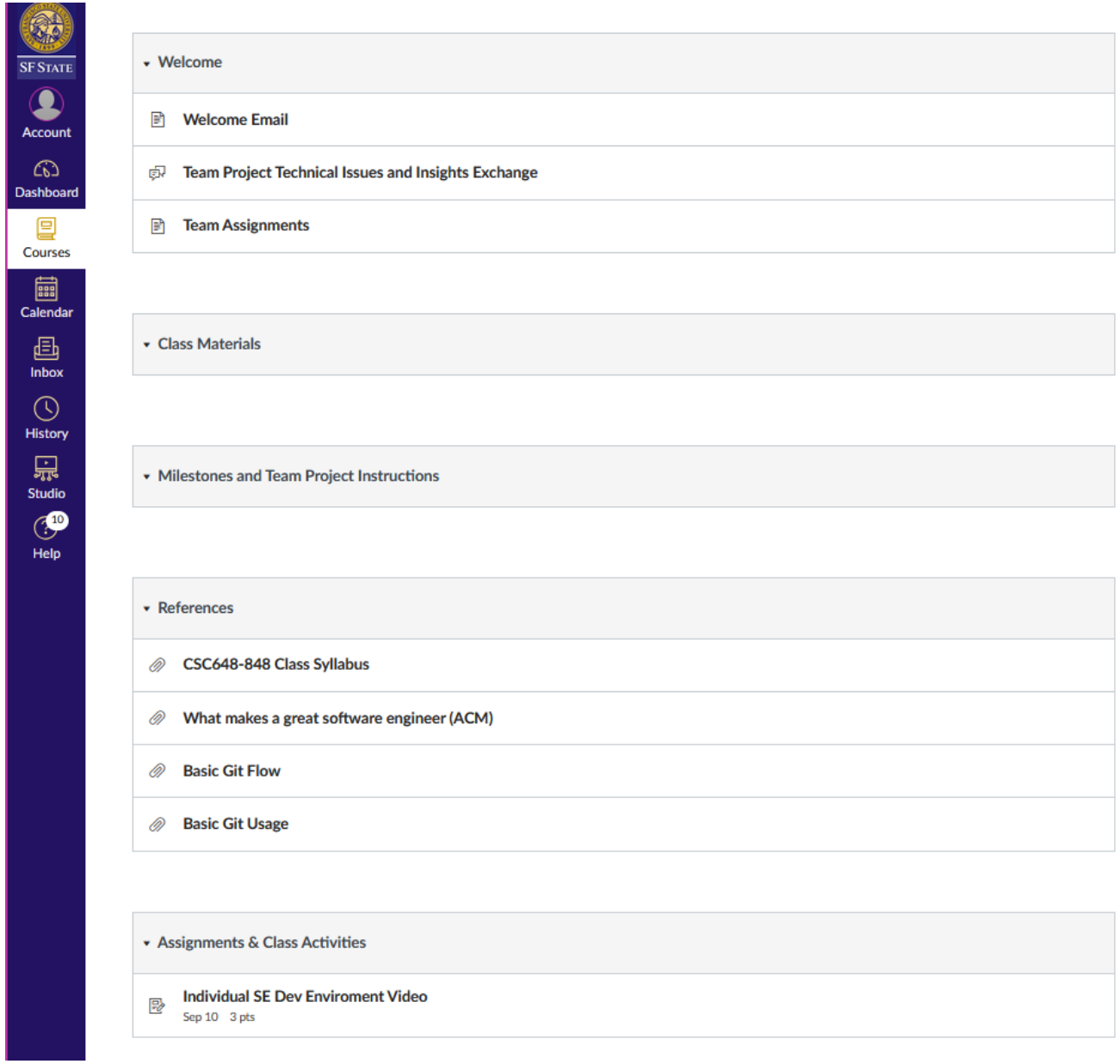
- Team project material and class slides will be posted on the SFSU Canvas ahead of time
- E-mail will be used by the instructors for general information and broadcasting to students. Will have "CSC648-848 section N team M" label in subject line (N is section number, M is team number).
- Must respond to emails from Instructor and from your team members and follow up in a timely manner (within 24 hours).
- If you send e-mail to instructors must have "CSC 648-848 section N Team M " in subject line (N class section, M is team number)
- Team leads communicate with instructors for general issues, then shares with the rest of the team
- Absence has to be communicated to team and instructor ahead of time
- Any delay in team delivery needs to be asked for via e-mail at least 24 h before the deadline

About the class – Grading

- Milestone 0 – individual assignment up to 10/100
- Team grade: student teams developing a product emulating small SW company, via Milestones 1-5 – 50/100. Demo and folder with documentation – M5 folder.
  - Team grade has *SE Product* component (how good is the product the team developed) and *SE Process* (how well the team followed SE best practices)
  - All team members get same grade unless exceptions (low attendance, low participation etc)
  - Team leads may get up to 5 points extra credit for job well done
  - Front end and back end team leads and github master my get up to 3 points extra credit for job well done
- Final exam: closed book, closed notes, in class/virtual. 40/100 of the grade (40 questions)

- No homework; No midterm BUT a lot of teamwork and 6 delivery milestones (M0-M5)

Canvas
https://sfsu.instructure.com



Class syllabus

- Check Canvas

Syllabus

Course Number: CSC 648/848
Course Title: SOFTWARE ENGINEERING/ADVANCED SOFTWARE ENGINEERING
Number of Credits: 3
Schedule: Three hours of lecture/discussion per week.
Prerequisite: a grade of C or better in CSC 413 and in CSC 317, or consent of the instructor

Catalog Description
Practical methods and tools for SW Engineering, including organizational teamwork. Course is paired with CSC 848. Students who have completed CSC 848 may not take CSC 648 for credit.

Expanded Description

- Introduction and motivation for Software Engineering (SE)
- Overview of several basic SE methodologies with emphasis on Iterative and Incremental Development and User Centered Design
- Usability and UI design principles and practice
- Basic components of SW Engineering process: Planning; Requirements and Specifications; Iterative Design, Rapid Prototyping, Mockups; Software Design; Coding and documentation techniques (high level only);
- SW Engineering related to Web application development
- Open source SW development and management
- Software Configuration Management, Code management, Delivery, Installation, and Documentation
- Software Metrics, Usability Measurements and evaluation
- Software QA and Testing
- Software Maintenance
- Project Management issues
- Teamwork and Communication as integral part of SW Engineering
- Issues related to global SW engineering
- Basics of IP, licensing, digital rights management and copyright
- SW Engineering ethics
- Real life examples and cases from instructor and students
- Guest and student presentations
- Final Group Project including several milestones, interaction with instructor, and final demo and
- delivery.

Course Objectives and Role in Program
This is a paired course (with graduate version CSC 848) which provides to the students an environment where they can integrate and apply all the knowledge they acquired in the program. It offers comprehensive and advanced coverage of practical methods and tools of full lifecycle SW development, as well as its organizational, teamwork and communicational aspects. Special emphasis will be on Iterative, Incremental, Agile and User Centered Design methodologies and on global SW engineering. Students will engage during the entire semester in a group project in order to experience and practice key aspects of SW engineering lifecycle in setting that simulates real SW company, and students play roles as back and front-end engineers, SW developers, tech leads and project managers. This course provides integrative experience in SW project organization and teamwork, including SW development, thus complementing student fundamental skills in computer science.

Learning Outcomes

- Knowledge of modern Agile SE engineering methods/processes and best practices for team-centric SW development, and their appropriate application
- Knowledge of basic components and tools for team-centric full lifecycle SW development from use cases and initial specifications to design, implementation, testing, evaluation as well as code management and deployment on cloud systems
- Knowledge of best practices for the design, implementation and evaluation for delivering easy to use and maintainable SW
- Ability to constantly iterate and re-prioritize plans and designs based on user needs, budget, schedule and resources
- Development of significant SE teamwork and project based experience, as close as possible to real life

Method of Evaluation

Individual assignment	10%	Students required to learn and install tools used in the team project
Final Group Project	50%	Design and development of a SW application by student teams. Grade is based both on adherence to SW Eng. Process, as well as on quality and usability of final project delivery in five deliverable milestones. Team graded as a whole
Final Exam	40%	Multiple choice questions testing basic knowledge of SW Engineering concepts and definitions

Graduate students may have to do a class presentation in order to get graduate credit for 848.

Final Team Project

Objective: By working on developing a SW application in a group setting, students will experience and exercise all key steps in full SW Engineering lifecycle simulating real SW company environment using the methodology of choice. Grading is based on adherence to SW Engineering methodology as well as the quality of the final product.

Organization: Students will work in teams of 5-6. Teams will be assigned by instructor based on internal selection, and then each team will assign themselves a team lead, front end lead, back end lead and github master. Instructor will serve as a coach, VP of Engineering, Marketing and Customer representative and will attempt to create environment that simulates real life SW company. Interaction with instructor will evolve around key SW Engineering process milestones and will include e-mail feedback, group walkthrough sessions and discussion, as well as usability feedback. Instructor will in general not get involved nor help in details of coding and implementation; this will be the responsibility of the students and the team.

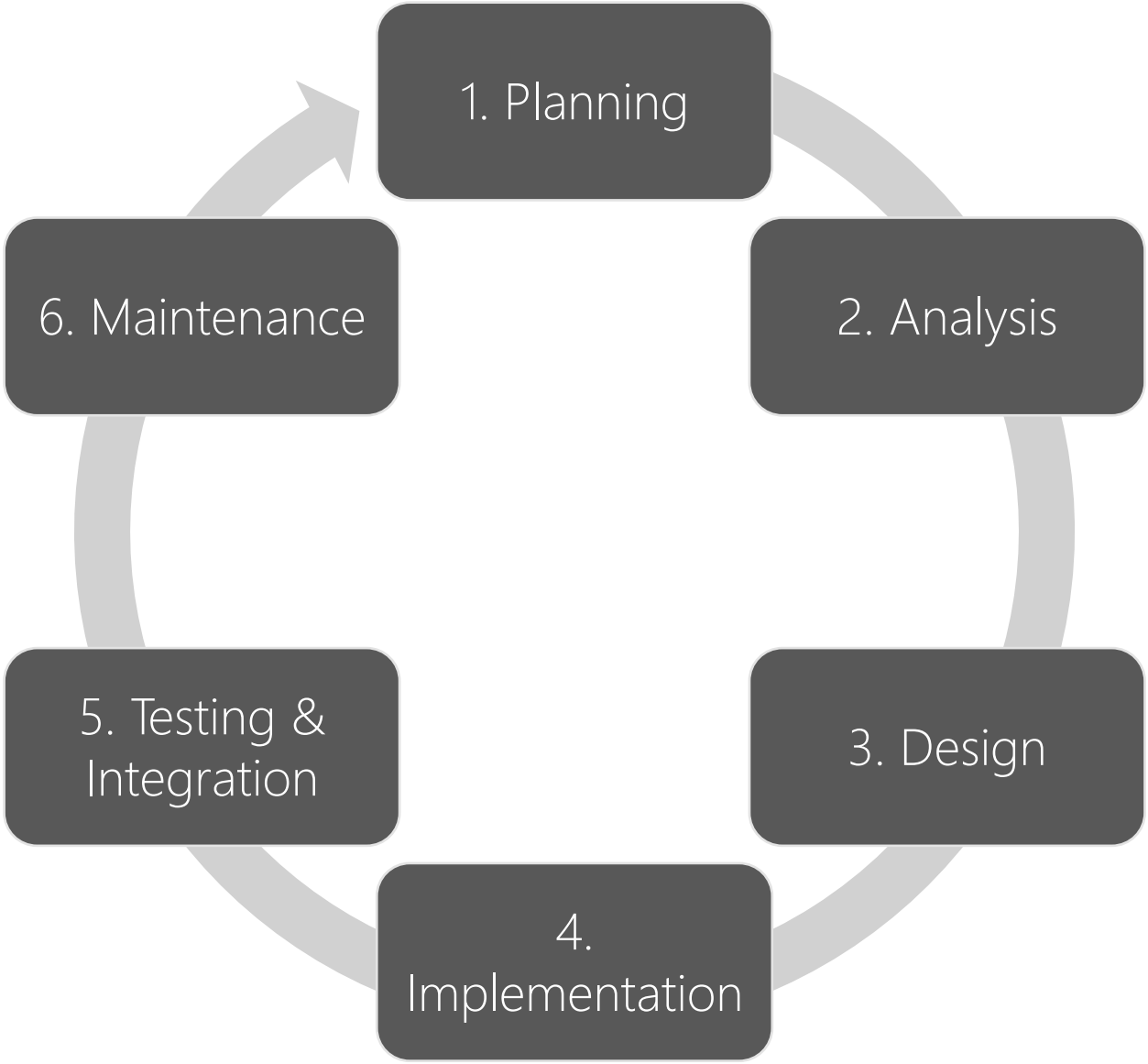
SW tools and IT environment: Basic SW tools and environment will be provided using open source SW and cloud platforms for deployment. Students will have some choice in deciding specifi SW stack and cloud systems from a set recommended by instructors.

Deliverables: These include a) Final documents from all the required milestones from the team project (after instructors' feedback has been incorporated), b) code and code management repository (e.g. github;

10 Minutes

Break

Software Development Life Cycle (SDLC) / SE Process Lifecycle



SW Engineering Process Components

1. *Requirements and Specification*
2. *Design and Implementation*
3. *Validation*
4. *Maintenance and Evolution*

Important

SW Engineering Process Components

- There are several basic components present in all SW Engineering methodologies:
  1. *SW Requirements and Specification*: functionality, constraints, look and feel, GUI, performance
  2. *SW Design and Implementation*
  3. *SW Validation*: formal, test plans, QA
  4. *SW Maintenance and Evolution*: ongoing changes (and fixes) to meet customer needs

  These components are applied in different way and order (linear, iterative, spiral…) depending on chosen SW Engineering methodology or models (I.e. *waterfall, extreme, agile* etc.)

Methodologies (Process Models)

- Waterfall
- Evolutionary
- Iterative
- Incremental
- Open Source Development
- Extreme Programing
- User Centered Design (UCD)
- Continuous Development
- AGILE & SCRUM

Important

SW Process Models (or Methodologies)

- Abstract representations of overall development of SW - methods and practices for organizing the SE effort
  - Waterfall
  - Evolutionary
  - Iterative
  - Open Source development
  - Extreme programming
  - For GUI and usability: User Centered Design (UCD) – variation of iterative development
  - Continuous development
- Agile, Scrum – high level project management (PM) processes used to "manage, organize and implement " the above SE processes

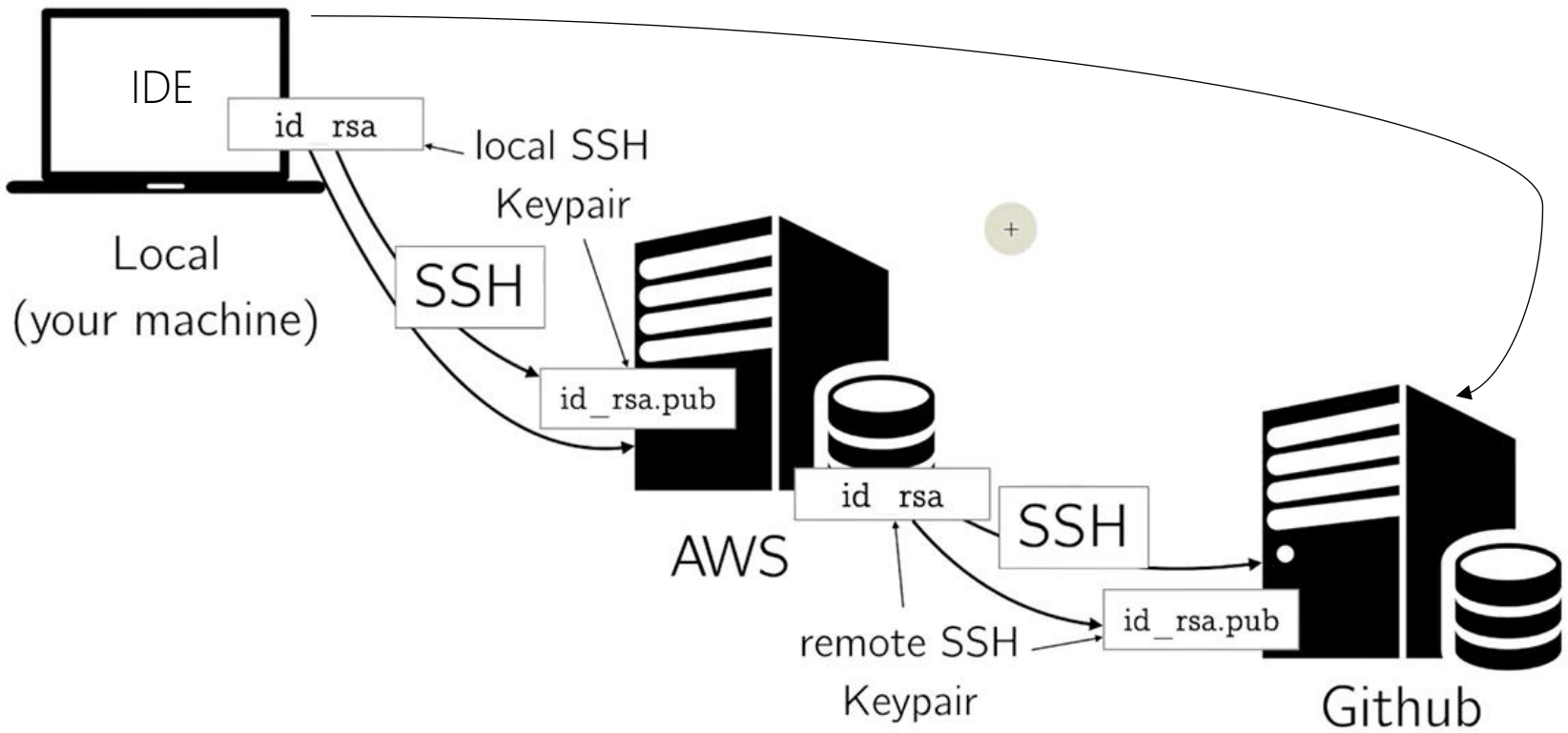
Milestone 0 – individual milestone – <u>Choose</u> and learn team project SW tools - 10/100

- First, students have to chose the tools and get approval from Instructor/CTO
- Then, they have to learn how to use chosen tools and create team page (ABOUT page)
- 10/100 of the grade -Details be presented and posted on Canvas – graded for individual team members separately
- Students within the team encouraged to help each other!
- Other benefits of M0:
  - Learn how to use tools for professional SW development
  - Learn basics of Cloud Computing and server management
  - Build teamwork

Class IT (for team project development)– Milestone 0

- Milestone 0 (M0) - Read Instructions
  - Class CTO recommends set of tools and frameworks
  - Student teams decide/propose what to use
  - Class CTO approves
  - Student install tools
  - Student build a team WWW page
- Team to chose/manage <u>their own delivery server</u> and <u>github</u>
- Teams will have one team lead/project manager and one front-end and one back-and leads instead of one CTO

Every single student should have this config
Part of the M0 (individual video)



Self Skill Survey

Member's Familiarity with Server-Side Language (Python, JavaScript, PHP) on a scale of 1 to 5, with 5 being very familiar and 1 being never used it.

- Jane: 5
- Jack: 2
- John: 3
- Joe: 1
- Jill: 4
- Jake: 5

Group/Team Milestones 1-5

- Class Team Project – High Level Description
- Milestone 1: High Level Req.&Specs, Use cases
- Milestone 2: More detailed specs, tools, servers and architecture defined, UI mock-ups, first vertical prototype
- Milestone 3: Project review (in-person): full horizontal (UI) and basic vertical prototype -feedback via focus group with Professor as customer "persona"; CTO reviews code and github off-line and sends feedback;
- Milestone 4: Beta delivery, QA plans and usability feedback, final functionality committed
- Milestone 5: Final Project demo ("Milestone 5 Folder") and demo of the SW – both graded for final team grade

Team Grading – 50/100

- All team members must contribute to the best of their abilities
- Team lead also participates in the development
- All group members get the same grade unless the instructors are told or discover that somebody is not contributing fully (e.g. poor attendance and communication, poor delivery on assigned tasks).
- Feedback and interaction are encouraged – hence the feedback, your questions, and intermediate project results are not graded. We grade only the final results after all the feedback is incorporated (M5 folder and final SW)
- All team leads may get up to 5 extra points; front-end and back-end team leads and github masters may get up to 3 points

Important

Participation and full focus on team project are required and are cornerstone of the class (50/100 points)

Team gets same grade for team project unless:

- Team lead reports (MUST be done as soon as problem occurs) or instructor discovers low participation e.g.
  - Not responsive to e-mails
  - Misses meetings with no excuse
  - Does not abide by schedules
- Instructor will investigate (e.g. speak to team lead, document, check github) then instructor will speak to individual and put him/her on formal notice
- Failure to improve may result from minimum 10% up to 50% of team grade reduction for the individual!!!!

Team grade -Final project grading – *same* grade per team

- Total 50/100 of the grade
  - Adherence and achievement of adequate *SE process* – **25%** (based on M5 folder and student behavior during the whole semester)
  - Quality of the delivered *SE product* – **25 %** (based on delivered SW and its architecture, gihub organization, code "smell;" etc.)

- All team members generally get the same grade
  - Rare Exception: In case of individual participation problems team grade for that individual shall be reduced

Rubrics for team project grading

Measured Team Outcomes for SE *process*	**Measured Team Outcomes for SE *product***
1. Fraction of the team participating at the meetings with the instructor 2. Quality and timing of follow-up on outstanding issues 3. Ability to deal with feedback constructively 4. Producing the non-SW and SW deliverables on time; following submission guidelines 5. Quality and completeness of non-SW deliverables (requirements, use cases, UI mockups, design documents, test plans etc.) as in M5 folder 6. Number of teamwork issues that instructor had to deal with (whether resolved or not) 7. Ability to apply best SE process and teamwork practices 8. Adherence to continuous and collaborative development practices 9. Ability to effectively use the SE tools (e.g. collaborative tools, version control, issue trackers)	1. Correctness and reliability of operation 2. Functionality of the product vs. desired requirements and use cases 3. Ease of use, user interface 4. Architecture of the developed system 5. Database design 6. Basic security, field validation 7. Performance 8. Code quality and “smell” (coding principles, style, documentation) 9. Proper usage of collaborative tools e.g. repository 10. Presentation style and effectiveness of final product demonstration (content, delivery, adherence to time, dealing with Q/A)
25 %	25 %

Final individual exam – 40% of the grade

- Modeled as a “certification test” – check knowledge of basic concepts and terms and application of best practices
- Five sections, 40 questions (each 1 point), 8 in each section:
  - SE Processes
  - Requirements and Specs
  - UI and usability
  - QA
  - Project Management
- Mainly multiple choice

Prep for final exam – what you need to learn

- Go over slides and remember class discussion
- Focus on main concepts, defections and best practices
- No need to memorize tools and technology names nor numbers (e.g. max typing speed etc.) - they are given in slides as reference material – they change often anyway
- For exam prep OK to only check slides marked Important

Important

Slides and Book (optional)

- NO book required
  - Class in part based on excellent SE book by **Prof. Ian Sommerville**, with additional material on WWW GUI and Usability from book by **McCracken**. The book is optional but recommended
  - Prof. Sommerville offers some material here:
    - https://iansommerville.com/software-engineering-book/static/videos/
    - We acknowledge copyright for the Prof. Sommerville slides we will use
- **Class slides** represent valuable material summarizing years of SE experience and best practices – keep them and use them at work.
- Both the book and slides are *of long term value* – they cover best practices and concepts which do not change often

How does the class prepare you for job market

- We simulate real world small company - comes as close to real work experience
- Your team project becomes your portfolio (M5 folder, WWW application and github)
- You improve your CV with the tools and SE processes you learned <u>and practiced</u>
- Instructions on how to improve your CV after the class are in http://cs.sfsu.edu/employment
  - See "Tips and Tricks…" slide

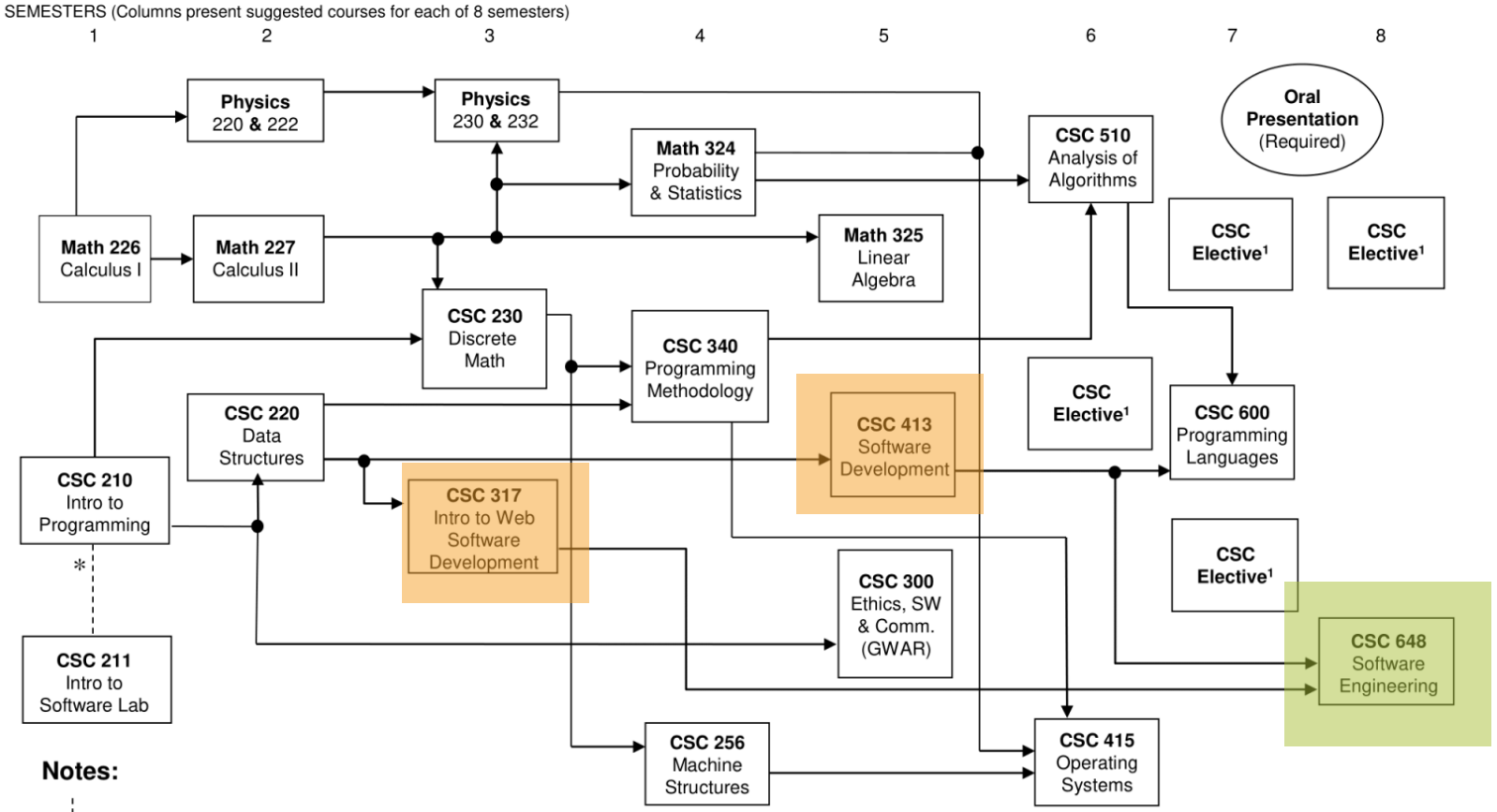
Prerequisites strictly observed

- Undergraduates
  - CSC 648: CSC 413 grade C or better (strictly enforced); no concurrent enrolment allowed
  - CSC 317 is also required.
- Conditionally classified grads who had CSC413 as a condition must clear this first, no concurrent enrolment allowed
- International visiting students - contact instructors

Pre-requisite chart

EFFECTIVE Fall 2019

SAN FRANCISCO STATE UNIVERSITY BACHELOR OF SCIENCE IN COMPUTER SCIENCE PREREQUISITE CHART



Notes:

* = Courses **MUST be taken concurrently**.

1. Four electives are required. Most courses above CSC 600 can be used as electives, as well as CSC 520 and MATH 400. (except CSC 600 and CSC 648 which is mandatory for major)
2. Check http://www.assist.org/web-assist/welcome.html to find equivalent courses at community colleges.

For more information and course descriptions visit http://cs.sfsu.edu

last updated: 04.04.19

Team Assignments

Last Name	First Name	Email	Team
Kumar	Mohit	mkumar3@sfsu.edu	1
Naranja	John Carlo	jnaranja@sfsu.edu	1
Sengchareune	Khaterina	ksengchareune@sfsu.edu	1
Hernandez	Alex	ahernandez75@sfsu.edu	1
Brust	Will	wbrust@sfsu.edu	1
Weerasuriya	Lasiru	lweerasuriya@sfsu.edu	1
Harish	Shreya	sharish@sfsu.edu	1
Sanchez	Brandon	bsanchez15@sfsu.edu	2
Wong	Laura	lwong25@sfsu.edu	2
Berhane	Mikias	mberhane1@sfsu.edu	2
Lubrino	Cielina Maree	clubrino@sfsu.edu	2
Le	Jason	jle34@sfsu.edu	2
Nadery	Omeid	onadery@sfsu.edu	2
Perez	Damian	dperez52@sfsu.edu	2
Romero	Martin	mromero13@mail.sfsu.edu	3
Velazquez Aguilera	Yanitsy	yvelazquezaguilera@sfsu.edu	3
Chen	Richie	lchen22@sfsu.edu	3
Teves	Kaylen	kteves@sfsu.edu	3
Rodriguez-Malak	Carlos	crodriguezmalak@sfsu.edu	3
Barra Avila	Franco	fbarraavila@sfsu.edu	3
Cancino	Cj	ccancino@sfsu.edu	3
Bhatta Lama	Sargam	sbhattalama@sfsu.edu	4
Ashby	Kaitlyn	kashby@sfsu.edu	4
Jezewski	Houdini	hjezewski@sfsu.edu	4
Torres	John	jtorres41@sfsu.edu	4
Wright	Emma	ewright12@sfsu.edu	4
Robles	Jacob	jrobles4@sfsu.edu	4
Qazi	Saeed	sqazi@sfsu.edu	4
Li	Mark	cli36@sfsu.edu	5
Jiang	Yuejun	yjiang13@sfsu.edu	5
Yu	Kerry	kyu19@sfsu.edu	5
Wahid	Mahvesh	mwahid@sfsu.edu	5
Kligman	Ian	ikligman@sfsu.edu	5
Kooner	Reman	rkooner@sfsu.edu	5
Caya	John Peter	jcaya@sfsu.edu	6
Rubio	Christopher	crubio4@sfsu.edu	6
Pasami	Gabriella	gpasami@sfsu.edu	6
Houlston	Foster	fhoulston@sfsu.edu	6
Young	Will	wyoung@sfsu.edu	6
Pokharel	Poojan	ppokharel1@sfsu.edu	6

SAN FRANCISCO STATE UNIVERSITY

Team Collaboration (8:00-8:45)

Get Familiar with your team

Review Final Team Project

Review M0

Outputs:

- List of Team Members Responsibilities/Roles
- SW stack chosen ready to ask for approval – see email template

Takeaways

Software Engineering (SE) is "The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software".

Most of the failures of delivering SW to specs, on time and budget, and to a user "delight" were rooted in a misunderstanding user needs, and/or poor design, planning and organization. SOFT SKILLS

Get familiar with the dynamic and organization of the class, calendar and milestones: SW stack approved by W2, Team Responsibilities by W2, and M0 is due in W3

Integrate fully with your team and collaborate

SE Process and Methodology

Plans for next Week

Team 1 present TAKEAWAYS from week 1 “SE Introduction & Motivation”

Week 1 Summary

Week 2 topic: Use Cases, Requirements and Specifications

Team Project, M0, and Team integration (lead/project manager, back/front-end lead, github master). Make sure you read the corresponding slides and the M0 documentation.

Enjoy the Learning Journey!